Floor Eijkelboom

Amsterdam eijkelboomfloor@gmail.com https://flooreijkelboom.github.io/ flooreijkelboom

About me

I am a **PhD researcher in generative AI** at the University of Amsterdam, working with Jan-Willem van de Meent and Max Welling. My research develops physics- and inference-inspired generative models that enhance controllability, creativity, and sample efficiency in large-scale AI systems. I introduced ***Variational Flow Matching (VFM)***, which frames diffusion and flow models through variational inference and has been applied across modalities, including molecules (proteins, materials), images, and discrete data. My current goal is to extend these methods beyond their present capabilities – enabling generative models to explore, compose, and innovate in structured and purposeful ways.

Education

PhD **University of Amsterdam**, Artificial Intelligence (Generative AI) — Sept 2024 – present

- PhD candidate in generative modeling at the University of Amsterdam, supervised by Jan-Willem van de Meent and Max Welling. Research on physics- and inference-inspired methods for principled and creative generative systems.
- Developed Variational Flow Matching (VFM), unifying diffusion and flow models through variational inference for stable, scalable, and controllable generation across diverse domains.

MSc **University of Amsterdam**, Artificial Intelligence (*cum laude*, with Honours) — 2022 – 2024

- Joint thesis between Oxford and Amsterdam, supervised by Erik Bekkers, Francesco di Giovanni, and Michael Bronstein, on topology and geometry in machine learning.
- Published three papers on topological and equivariant deep learning; co-lecturer for the MSc Machine Learning course.

BSc **Utrecht University**, Artificial Intelligence (*cum laude*, with Honours) — 2019 – 2022

- Minored in pure mathematics and completed the philosophy honours track.
- Chaired the curriculum committee and was finalist for "Best Representative of Utrecht University 2019/2020."

Teaching Experience

Visiting Lecturer & Co-Organizer, Generative Modeling Course (African Institute for Mathematical Sciences) — Cape Town, South Africa, 2025

- Co-organized and lectured a graduate-level course on generative modeling at the African Institute for Mathematical Sciences (AIMS), a pan-African network for post-graduate training in mathematical sciences.
- Designed lectures and labs introducing flow matching, diffusion models, and variational inference through physics- and inference-inspired perspectives.
- Developed hands-on sessions connecting probabilistic inference with physics-based generative modeling.

Lecturer, Machine Learning 1 (MSc Artificial Intelligence, University of Amsterdam) — Amsterdam, Netherlands, 2023 – present

- Teach a core MSc course introducing the mathematical and probabilistic foundations of modern machine learning.
- Design and deliver lectures, tutorials, and assignments on inference, optimization, and representation learning.
- Mentor MSc students, leading to multiple papers.

Publications

Riemannian Variational Flow Matching for Material and Protein Design 2025
O. Zaghen, ***F. Eijkelboom***, A. Pouplin, C. Liu, M. Welling, J.-W. van de Meent, E. J. Bekkers
Arxiv Preprint

Purrception: Variational Flow Matching for Vector-Quantized Image Generation 2025
R.-A. Matișan, V. T. Hu, G. Bartosh, B. Ommer, C. G. M. Snoek, M. Welling, J.-W. van de Meent, M. M. Derakhshani, ***F. Eijkelboom***
Arxiv Preprint

Discovering Lie Groups with Flow Matching 2025
J. Y. Park, Y. Chen, ***F. Eijkelboom***, J.-W. van de Meent, L. L. S. Wong, R. Walters
Arxiv Preprint

Controlled Generation with Equivariant Variational Flow Matching 2025
F. Eijkelboom, H. Zimmermann, S. Vadgama, E. J. Bekkers, M. Welling
International Conference on Machine Learning (ICML 2025)

Exponential Family Variational Flow Matching for Tabular Data Generation 2025
A. Guzmán-Cordero*, ***F. Eijkelboom****, J.-W. van de Meent
International Conference on Machine Learning (ICML 2025)

Towards Variational Flow Matching on General Geometries 2025
O. Zaghen, ***F. Eijkelboom***, A. Pouplin, E. J. Bekkers
ICLR 2025 Workshop on Deep Generative Models in Machine Learning: Theory and Practice

Variational Flow Matching for Graph Generation 2024
F. Eijkelboom*, G. Bartosh*, C. Andersson Naesseth, M. Welling
Advances in Neural Information Processing Systems (NeurIPS 2024), 11735–11764

E(n) Equivariant Message Passing Cellular Networks 2024
V. Kovač, E. J. Bekkers, P. Liò, ***F. Eijkelboom***
ICML 2024 Workshop on Geometry-grounded Representation Learning

Clifford Group Equivariant Simplicial Message Passing Networks 2024
C. Liu*, D. Ruhe*, ***F. Eijkelboom***, P. Forré
International Conference on Learning Representations (ICLR 2024)

Can Strong Structural Encoding Reduce the Importance of Message Passing? 2023
F. Eijkelboom, E. J. Bekkers, M. M. Bronstein, F. Di Giovanni
Topological, Algebraic and Geometric Learning Workshops (TAGL 2023), 278–288

E(n) Equivariant Message Passing Simplicial Networks 2023
F. Eijkelboom, R. Hesselink, E. J. Bekkers
International Conference on Machine Learning (ICML 2023), 9071–9081